F232/3 Multi-Axis Loadcell

The F232 measures forces in two axes at 90° and the F233 measures forces in 3 mutually perpendicular axes. Apart from error evaluations, each output is pure and requires no mathematical manipulation.

The F232 is designed to measure forces in two perpendicular directions, while the F233 captures forces along all three orthogonal axes. Each channel provides an independent output, allowing measurements to be taken without the need for mathematical cross-correction, aside from standard error considerations.

As with many 3 Axis Load Cells, these units are moment-sensitive, meaning calibration must be performed at a defined force centre. The standard reference point is listed in the specification, but alternative centres can be supplied for applications that require them. Our engineering team can assist with determining the most suitable calibration method for specialized setups.

These multi-axis load cells are well-suited to a wide range of industrial and research environments, including work within automotive development, medical engineering, and other fields where simultaneous directional force measurement is critical. The devices can be produced with force capacities tailored for specific projects, and the Z-axis range can be configured independently from the X and Y axes. The illustrated model represents a 500N

Features

- 2 & 3 axis versions
- Custom force ranges
- Simple installation
- Direct output from each axis without calculation
- Traceable calibration with certificate included in the standard price
- Standard 1 year warranty

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version; dimensions may vary slightly for other load ranges.

For projects needing higher capacity 3 Axis Load Cells or alternative multi-axis configurations, Novatech can manufacture custom variants with ranges up to 20kN. High-temperature options are also available for environments requiring elevated operational limits.

Additional guidance on selecting and specifying multi-axis load cells is provided in Engineering Sheet E015, which offers useful information for engineers determining the most appropriate configuration for their system.

Whether for laboratory testing, field measurement, or integration into OEM assemblies, the F232 and F233 combine precision outputs with highly adaptable design options, making them dependable solutions for complex force analysis requirements.

FAQ’s:

How do I know if the F232 or F233 is right for my application?

The F232 measures force in two axes, while the F233 measures in three. If your application requires simultaneous X-Y or X-Y-Z force insights, these sensors are suitable. Our engineers can advise based on load ranges, mounting limits, and measurement goals.
Can Novatech calibrate the load cell for a non-standard force centre?

Yes. Since these 3 Axis Load Cells are moment-sensitive, we can calibrate them at alternative force centres when required. Our engineering team will help determine the most appropriate calibration point for your setup to ensure accurate multi-directional measurements.
Are custom force ranges or high-temperature versions available?

Absolutely. We can manufacture the F232 or F233 with tailored force capacities, including different values for the Z-axis. Custom variants up to 20kN are available, along with high-temperature options for demanding industrial or research environments.

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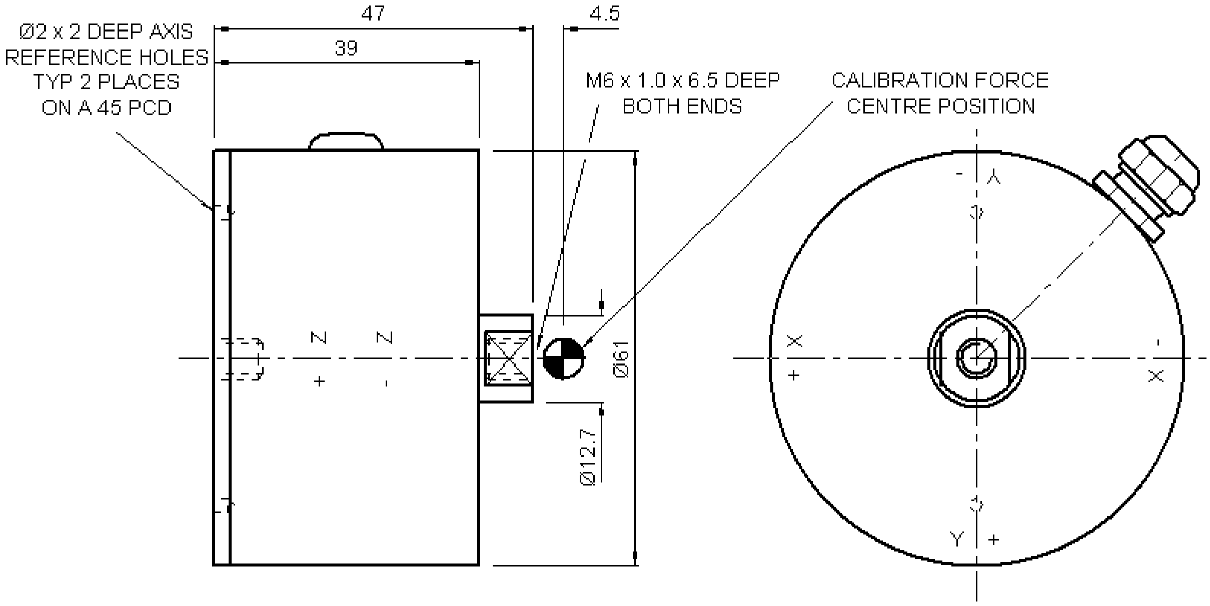
Outline



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Specifications

Parameter	Value	Unit
Non-linearity - Terminal	±0.5	% RL
Hysteresis	±0.5	% RL
Creep - 20 minutes	±0.1	% AL
Repeatability	±0.02	% RL
Maximum cross talk	3	% RL
Calibration force centre	x=0, y=0, z=-4.5	mm
Rated output - Nominal	1.2	mV/V
Zero load output	±4	% RL
Temperature effect on rated output per °C	±0.005	% AL
Temperature effect on zero load output per °C	±0.01	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	10	V
Bridge resistance X & Y axis	350	Ω
Z axis	700	Ω
Insulation resistance - Minimum at 50Vdc	500	MΩ
Overload - Safe	50	% RL
Overload - Ultimate	100	% RL
Weight - Nominal (excluding cable)	200	g

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Order Codes

Code	Description
Most F232/3 loadcells are manufactured to special requirements and are given an F232-Zxxxx or F233-Zxxxx number.	

Structural Stiffness

Range (kN)	Stiffness (N/m)
100 N (per axis)	
5 kN (per axis)	

Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.
- Values apply to all axes unless otherwise specified.

Connections

The F232 is fitted with 2 metres of PVC insulated 9 core screened cable type 7-1-9C. The F233 is fitted with 2 metres of PVC insulated 12 core screened cable type 7-1-12C. The screen is not connected to the loadcell body.

Function		Wire Colour	
	X axis	Y axis	Z axis
Excitation +	Red	Violet	Orange
Excitation –	Blue	Black	Turquoise
Signal +	Yellow	Brown	Pink
Signal –	Green	White	Grey
Screen	Orange (thick)		

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